



WPI

Building AI-Enhanced Educational Websites

A Hands-On Workshop for Engineering Educators

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Workshop Objectives

1

Apply Vibe Coding

Use ChatGPT as your coding partner to prompt, iterate, and refine*

2

Build & Publish a Website

Create and host a real educational site using GitHub Pages

3

Brand Your Site

Match your university's colors, fonts, and visual identity

4

Design Responsibly

Keep sensitive information safe, document your workflow, maintain ethical transparency



**You can replace ChatGPT in this presentation with any other comparable AI model.*



What is Vibe Coding?

PROMPT



Describe what you want
in plain language

GENERATE



ChatGPT writes the code
instantly

INSPECT



Read, tweak, and test
in your browser

ITERATE



Refine by conversation,
not by expertise



Where We're Headed Today

- You will leave with a live, published website
 - Branded to your institution's visual identity
 - Built entirely through prompting ChatGPT — no prior coding needed
 - Ready to extend with resources, links, and (optionally) AI features
-



studyguideai.app

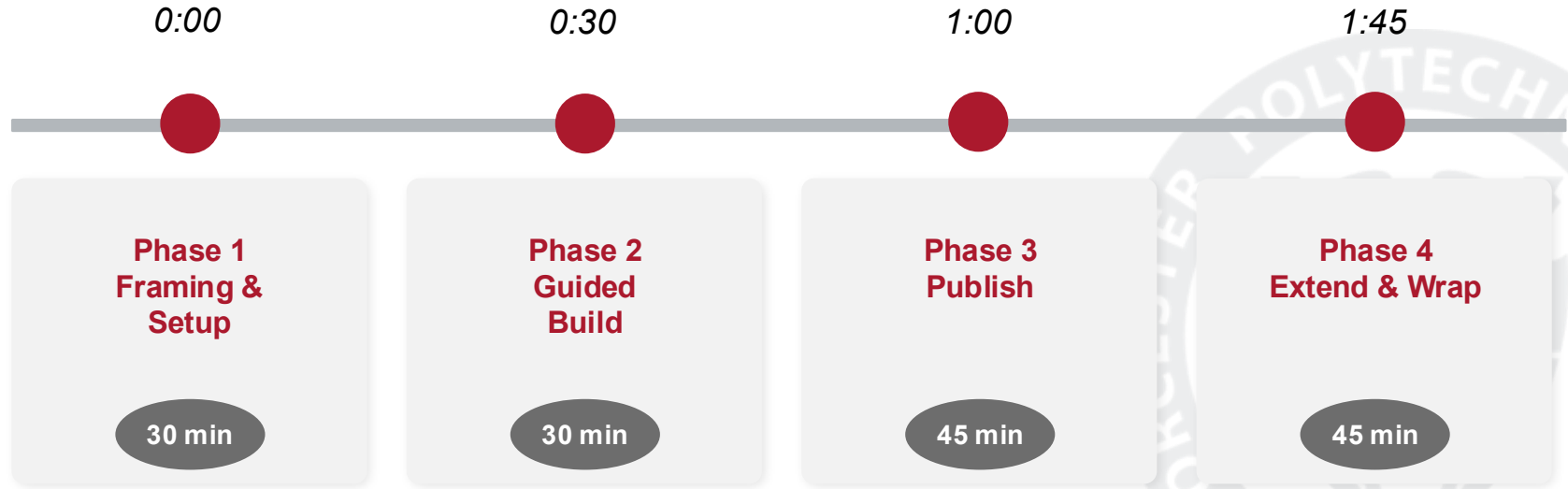


aipedagogy.wpi.edu



gpthermo.wpi.edu

Workshop Agenda — 150 Minutes



Note: Fast-laners can help neighbors. Everyone leaves with a live URL.

What Makes a Website?

Structure & Content

`index.html`

- Describes **what** is on the page and how it is organized
 - (headings, paragraphs, images, links)
- Human-readable, inspectable, editable by anyone.

ChatGPT writes this. You review it.

Style & Appearance

`css/styles.css`

- Controls **how** everything looks
 - (colors, fonts, spacing, and layout)
- Kept separate so changing your brand color means editing **one line**, not every element.

ChatGPT writes this too.

Assets

`images/ files/`

- Files the site references but does not generate
 - (photos, logos, PDFs, documents)
- Keeping them in named folders means swapping a photo or syllabus never breaks the HTML.

*You provide these.
ChatGPT just references them.*

Your Project Folder

```
my-course-site/  
├── index.html  
├── css/  
│   └── styles.css  
├── images/  
│   ├── logo.png  
│   └── photo.jpg  
└── files/  
    └── syllabus.pdf
```

✓ Use this

```
index.html  
css/  
styles.css  
images/  
templates/
```

✗ Avoid this

```
my-page.html  
styling/  
design.css  
pics/  
pages/
```

Convention is communication

Debugging is faster

When something breaks, every tutorial (e.g. Stack Overflow answer) assumes these names — solutions apply directly

ChatGPT recognizes standard names instantly

Trained on millions of projects using `index.html`, `css/`, `images/` — it knows exactly what you mean

Servers expect `index.html` by default

GitHub Pages, Netlify, and every web host serves `index.html` automatically — no configuration needed

Idiosyncratic names create ambiguity that costs you tokens, time, and patience

Extracting Your Institution's Brand

The 60-30-10 Color Rule

60% Dominant: white/light backgrounds

30% Secondary: brand color, headers, nav

10% Accent: borders, highlights, footer

WPI example:

- 60% #FFFFFF white
- 30% **#AC2B37** crimson
- 10% **#A8AAAD** grey

You only need 3 colors — ignore the rest

Industry Connections Amplify Our Graduate Programs

Grab the momentum and grow your career.

[Why WPI for Grad School?](#)

Making the Most Successful College Transition

The newest WPI Podcast offers practical and proven tips to get the first year off to a strong start.

[Listen to the WPI Podcast](#)

The AI-Driven World Cup

How will data, augmented reality, and other tech transform this year's tournament?

[Ask Our Experts](#)

A Degree That Matters

At WPI, you'll use your unmatched college experience—built around STEM, business, global engagement, and the humanities—to work in teams here at home and around the world to create solutions to problems faced by real people.

Explore the possibilities with more than 70 undergraduate and graduate degree programs across [four schools: Engineering, Arts & Sciences, Business and Global](#).

FIND A DEGREE PROGRAM:

BACHELOR'S

MASTER'S

PHD

Latest News

[> SEE MORE NEWS](#)



New Worcester Polytechnic Institute Research Could Give Used Electric Vehicle Batteries a Second Life as Higher-Performance Materials



WPI

Extracting Your Institution's Brand

3-Step Brand Extraction

Write down your 3 color hex codes

1

Screenshot your homepage

Captures layout feel & secondary colors



2

Inspect for exact hex codes

Right-click → Inspect → find color in CSS panel

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FIND A DEGREE PROGRAM:

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Latest News

> SEE MORE NEWS



New Worcester Polytechnic Institute Research Could Give Used Electric Vehicle Batteries a Second Life as Higher-Performance Materials

3

Prompt ChatGPT with both

Screenshot + exact hex = accurate, intentional design



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Gather Your Files

1 Your photo
Rename it to photo.jpg (It's OK if you have a different extension)

2 Institution's logo
Rename it to logo.png (It's OK if you have a different extension)

3 Course syllabus
CourseName_syllabus.pdf (It's OK if you have a different name and/or extension)

4 Project folder
Set up your project folder

```
my-course-site/  
├── index.html  
├── css/  
│   └── styles.css  
├── images/  
│   ├── logo.png  
│   └── photo.jpg  
└── files/  
    └── syllabus.pdf
```

Let the Vibes Begin

I am building a single-page faculty course hub website.

Generate two files: **index.html** and **css/styles.css** linked via relative path.

Folder structure to follow exactly:

```
my-course-site/
├── index.html
├── css/
│   └── styles.css
├── images/
│   ├── logo.png
│   └── photo.jpg
└── files/
    └── syllabus.pdf
```

My information:

- Name and title: **[YOUR NAME AND TITLE]**
- Institution: **[YOUR UNIVERSITY NAME]**
- Course: **[COURSE NUMBER AND NAME]**
- Primary brand color: **[HEX]** — use for nav bar, section headers, footer
- Secondary brand color: **[HEX]** — use for borders, secondary text



[Scan for the starter prompt](#)

Let the Vibes Continue

Page sections:

- Nav bar with images/[logo.png](#) on the left, course title on the right — primary color background, white text
- Hero section — images/[photo.jpg](#) as a round-cropped photo on the left, name, title, and [one-sentence bio](#) on the right
- Course description — white card with primary-color left border, [two short placeholder paragraphs](#)
- Resources — three cards side by side: Syllabus linking to files/[syllabus.pdf](#), Office Hours [add office hours](#), Contact info [add contact info](#)
- Footer — primary color background, white text, centered copyright line

Rules:

- All styling in `css/styles.css` — no inline styles
- Flexbox layout, responsive at 760px breakpoint
- No JavaScript, no external libraries, no frameworks
- Must work by double-clicking `index.html` — no server needed
- Add a comment at the top of `styles.css` listing your brand colors and their roles



[Scan for the starter prompt](#)

How Are The Vibes?

1. Scan the QR code for the starter prompt
2. Customize the prompt
3. Submit to ChatGPT
4. Download the index.html and styles.css
5. Make sure you have the proper project folder hierarchy
6. Double-click index.html (or open with browser)
7. Inspect and modify it if necessary (**I'll demo this**)



[Scan for the starter prompt](#)

Let The World Know Why GitHub?

It is not just hosting



Version Control

- Every upload is a saved snapshot.
- Made a mistake? Roll back to any previous version.

Your site has an undo button



Cloud Backup

- Your code lives in the cloud, not just on your laptop.
- Laptop stolen or crashes; your site is safe.

Your site survives your hardware



Publishing Pipeline

- GitHub Pages turns your repo into a live website automatically.
- Upload → live URL. Free, forever.

username.github.io/your-repo

Use **Netlify.com** if GitHub doesn't work out (publish in a minute!)

GitHub Setup

1. If you already have a GitHub account, then you are set for this step.
 - Otherwise, go to github.com and create an account.
2. Login and create a repository

New repository

Type to search

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *
rebadiwpi

Repository name *
course-hub
✓ course-hub is available.

Description
Prof. Ebadi's course hub
24 / 350 characters

2

Configuration

Choose visibility *
Choose who can see and commit to this repository
Public

Add README
READMEs can be used as longer descriptions. [About READMEs](#)
Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)
No .gitignore

Add license
Licenses explain how others can use your code. [About licenses](#)
No license

Create repository

GitHub Setup

1. Upload the material

To create a folder:

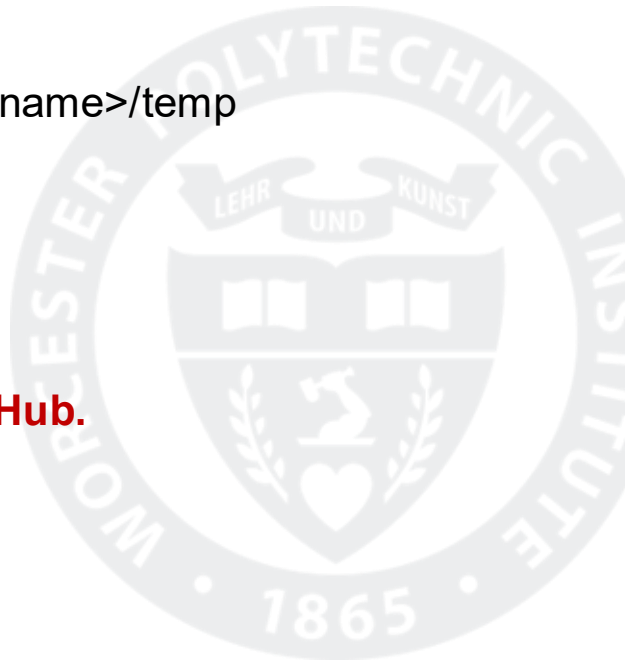
- Click **Add files** (then create new file), then add <folder name>/temp
 - e.g. css/temp
- Upload styles.css, then delete temp

2. Repeat for all files and folders

Don't publish sensitive data (APIs, student info, etc.) to GitHub.

3. Publish the website

- Go to **Settings**, then navigate to **Pages**
- Choose **main** branch and wait 1-2 minutes



Extend The Vibes

Prompt ChatGPT to change the index.html to include additional courses

- Let's make some changes. I want to make this website a hub for all my courses instead of one. Update the header accordingly.
- Here are my courses: ME 3310 Kinematics of Mechanisms, ES 3001 Thermodynamics, ES 1310 Introduction to CAD.
- Replace the existing course description section with one box per course.
- Each box should have two links: a syllabus and a lecture file, both located in the **files/folder**, using this naming format: course_number_syllabus.pdf and course_number_lecture.pptx
- for example files/me3310_syllabus.pdf and files/me3310_lecture.pptx.

Extenddd The Vibes

Prompt ChatGPT to organize the website so each course has its own html page:

- Now create a new HTML page for each course inside the **templates/** folder.
- On the home page, make each course box link to its respective course page.
- Each course page should include a resources section with links to the syllabus and lecture files using `../files/` as the path prefix
- for example `../files/me3310_syllabus.pdf`.
- Each course page should also include a navigation link back to the home page.

Let's Keep The Vibes Up



studyguideai.app



aipedagogy.wpi.edu



gpthermo.wpi.edu

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